Erfan Hajihashemi

erfan.hajihashemi@gmail.com LinkedIn (669) 241-9744 Google Scholar

Education

University of California, Irvine *Sept. 2023 – Present* | **Irvine, USA**
Ph.D. in Electrical Engineering and Computer Science, GPA: 3.96/4, ***M.Sc. conferred June 2025***

University of Tehran *Sept. 2018 – May 2023* | **Tehran, Iran**
B.Sc. in Electrical Engineering, Minor in Computer Engineering, ***Best Undergraduate Thesis Award***

Research Interests

Trustworthy AI, LLM Reasoning and Verification, Online Learning

Research Experience

University of California, Irvine *Sept. 2023 – Present* | **Irvine, CA**
Graduate Research Assistant | ***Advisor: Prof. Yanning Shen***

- Proposed **OMVV**, an online **LLM verification** algorithm that casts verifier selection as a **bandit** problem, routing each response among weak verifiers and escalating to a costly oracle only when needed, achieving higher accuracy at lower cost than any fixed verifier.
- Developing a framework that leverages **conformal prediction** to provide statistically guaranteed response selection for **LLM reasoning**, employing principled uncertainty quantification to mitigate **hallucination** and balance assertiveness and conservativeness in multi-step reasoning tasks.
- Collaborating with a neuroscience team at UC Irvine on **time-series prediction** of cortical neural oscillations from hippocampal signals, incorporating **explainable AI** techniques for interpretable feature selection.
- Proposed **SAMOCP**, a multi-model ensemble conformal prediction framework for dynamic environments with distribution shifts; proved strongly adaptive regret guarantees over arbitrary time intervals while maintaining valid coverage, advancing **reliable uncertainty quantification** for trustworthy machine learning.
- Proposed **GMOCP** and **EGMOCP**, multi-model online conformal prediction algorithms that construct and refine a **bipartite graph** at each time step to dynamically prune weak models and identify effective model subsets, yielding smaller and more **reliable prediction sets** with provable coverage guarantees.

Publications

- **Erfan Hajihashemi**, Yanning Shen, "Online Verification of Language Model Responses Under Cost Constraints," Under review (arXiv), 2026.
- **Erfan Hajihashemi**, Yanning Shen, "Multi-model Ensemble Conformal Prediction in Dynamic Environments," Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2024.
- **Erfan Hajihashemi**, Yanning Shen, "Multi-model Online Conformal Prediction with Graph-Structured Feedback," Transactions on Machine Learning Research (**TMLR**), 2025. **Awarded J2C certification**.
- **Erfan Hajihashemi**, Yanning Shen, "Enhanced Multi-model Online Conformal Prediction," Asilomar Conference on Signals, Systems, and Computers (**Asilomar**), 2025.

Technical Skills

- **Programming Languages:** Python, C++, C, R, LaTeX
- **Scientific Libraries:** PyTorch, TensorFlow, Keras, Transformers, Scikit-learn, NumPy, Pandas, CUDA

Honors & Awards

- Awarded 2nd place in the Annual Tech Showcase, EECS Department, University of California Irvine
- Henry Samueli Endowed Fellowship, 2025-2026 academic year
- EECS Ph.D. Fellowship, University of California Irvine, 2023-2024 academic year
- Ranked 211th in the National Universities Entrance Examination among more than 150,000 contestants, 2018

Reviewer

- NeurIPS, ICLR, TMLR